Smoothed Particle Hydrodynamics Study Notes

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1 Fluid Simulation Methods and Where SPH Fits

In numerical fluid simulation, continuous fields such as velocity, density, pressure, and temperature must be represented by finitely many degrees of freedom. Methods are often classified by how those degrees of freedom represent space and how they move relative to the fluid. The broadest distinction is between **Eulerian** methods, which observe fluid through fixed locations, and **Lagrangian** methods, which follow moving parcels of material. Hybrid methods combine both viewpoints.

1.1 Eulerian Mesh-Based Methods

Eulerian methods divide the spatial domain into a structured or unstructured mesh and store physical quantities at grid nodes, cell centers, or element faces. Fluid flows through this fixed spatial representation. Common examples include:

- **Finite Difference Methods (FDM):** Approximate derivatives directly from values on a usually structured grid. They are conceptually simple and efficient, but are less convenient for complicated geometries.
- **Finite Volume Methods (FVM):** Integrate the governing equations over control volumes and compute fluxes through cell faces. Their local conservation properties make them especially common in computational fluid dynamics (CFD).
- **Finite Element Methods (FEM):** Represent the solution with basis functions over mesh elements. They naturally support irregular geometries, variable resolution, and rigorous weak formulations.
- **Lattice Boltzmann Methods (LBM):** Evolve discrete particle distribution functions on a lattice rather than discretizing the Navier–Stokes equations directly. They are highly parallel and useful for complex or porous boundaries, although standard formulations use regular lattices and have restrictions on compressibility and stability.

Mesh-based methods generally provide accurate differential operators, mature pressure solvers, and efficient computation on regular grids. They are often preferred for flows in fixed domains and for high-accuracy engineering CFD. Their main difficulty is that the fluid must be advected through the grid, which can introduce numerical diffusion. Free surfaces require an additional interface representation, such as a level-set or volume-of-fluid method. A fixed Eulerian mesh itself does not deform, but moving boundaries and interfaces can still require cut-cell, immersed-boundary, adaptive-mesh, or remeshing techniques.

1.2 Lagrangian and Mesh-Free Methods

Lagrangian methods move their computational degrees of freedom with the material. **Smoothed Particle Hydrodynamics (SPH)** is both a Lagrangian particle method and a mesh-free method: it represents the fluid by particles without permanent connectivity between them. Each particle carries quantities such as mass, position, velocity, density, and pressure. Field values and their derivatives are estimated from weighted sums over neighboring particles using a smoothing kernel.

Other methods in or near this category include **vortex methods**, which represent vorticity using moving particles or filaments, and other mesh-free collocation or moving-least-squares methods. These methods differ from SPH in what their particles represent and how they reconstruct continuous fields.

1.3 Hybrid Particle–Grid Methods

Particle-in-cell methods use particles to transport material information and a background grid to compute forces or solve pressure. Important variants include **PIC**, **FLIP**, and **APIC**. The **Material Point Method (MPM)** uses a related particle–grid transfer for fluids, solids, and coupled materials. These approaches seek a compromise: particles handle advection and changing topology, while the grid supplies structured and often more stable differential operators. Their disadvantages include transfer error between particles and the grid, grid-crossing artifacts, and the complexity of maintaining two representations.

1.4 Benefits of SPH

- **Natural free surfaces:** A fluid boundary is simply the edge of the particle distribution, so no separate interface-tracking field is required for many applications.
- **Large deformation and topology changes:** Splashing, fragmentation, merging, and moving boundaries do not distort a mesh or require connectivity to be rebuilt.
- **Lagrangian advection:** Material quantities move with the particles, reducing the numerical diffusion associated with transporting them through a fixed grid.
- **Conservation and coupling:** Symmetric pairwise force formulations can conserve mass exactly and conserve linear and angular momentum well. The particle representation also couples naturally to moving rigid bodies and deformable solids.
- **Locality and parallelism:** Interactions have compact support and are mostly local, making SPH suitable for parallel CPU and GPU implementations.

1.5 Limitations of SPH

- **Computational cost:** Neighbor searches must be updated as particles move, and a well-resolved three-dimensional simulation can require many particles and interactions.
- **Accuracy and consistency:** Kernel approximations are generally less accurate per degree of freedom than high-order mesh methods. Irregular or incomplete particle neighborhoods, especially near boundaries, can reduce consistency and introduce error.
- **Pressure and density errors:** Weakly compressible SPH permits small density fluctuations and may require very small time steps because of an artificial speed of sound. Incompressible SPH variants instead require a more expensive global pressure solve.
- **Stability artifacts:** Poor formulations can suffer from tensile instability, particle clumping, noisy pressure fields, or excessive artificial viscosity. Stabilization improves robustness but can add dissipation or alter the modeled physics.
- **Boundary treatment:** Solid walls are not automatically represented accurately. Ghost particles, boundary particles, or corrected kernels are often needed to impose no-penetration and no-slip conditions.
- **Resolution and convergence:** Refinement, adaptive resolution, and rigorous convergence studies are more difficult than in many established mesh-based schemes.

SPH therefore occupies an important but application-dependent part of the fluid-simulation field. It is particularly attractive when free surfaces, moving geometry, violent motion, or fluid fragmentation dominate the problem. For smooth flows in fixed domains, boundary-layer accuracy, or simulations requiring high accuracy at lower computational cost, finite volume, finite element, or high-order finite difference methods are often more appropriate. Hybrid particle–grid methods are frequently chosen when particle-based advection is desirable but a grid-based pressure solve or derivative operator offers better stability.

2 Context in Fluid Simulation: Functions and Partial Differential Operators

Fluid motion is governed by continuous continuum mechanics equations, primarily the **Navier–Stokes equations** for momentum conservation and the **continuity equation** for mass conservation.

2.1 Governing Equations

For a standard single-phase Newtonian fluid, the governing model consists of conservation laws for mass and momentum, constitutive and closure relations, and appropriate initial and boundary conditions. Thermal flows additionally require an energy equation. Multiphase, reactive, viscoelastic, or magnetohydrodynamic simulations require further equations beyond those listed here.

2.1.1 Lagrangian Description and Material Derivative

Because SPH particles move with the fluid, SPH naturally uses the Lagrangian form of the equations. The material derivative follows a moving fluid parcel:

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla. \tag{1}$$

Particle trajectories satisfy the kinematic equation

$$\frac{D\mathbf{x}}{Dt} = \mathbf{v}. \tag{2}$$

2.1.2 Conservation of Mass

The continuity equation in Lagrangian form is

$$\frac{D\rho}{Dt} = -\rho(\nabla \cdot \mathbf{v}) \tag{3}$$

or, equivalently,

$$\frac{D\rho}{Dt} + \rho\nabla \cdot \mathbf{v} = 0. \tag{4}$$

For an incompressible fluid of constant density ρ_0, this reduces to the kinematic constraint

$$\nabla \cdot \mathbf{v} = 0, \qquad \rho = \rho_0. \tag{5}$$

In SPH, density may be evolved with the continuity equation or recomputed using the summation density $\rho_i = \sum_j m_j W_{ij}$.

2.1.3 Conservation of Linear Momentum

The Cauchy momentum equation is

$$\rho \frac{D\mathbf{v}}{Dt} = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{b}, \tag{6}$$

where $\boldsymbol{\sigma}$ is the Cauchy stress tensor and $\mathbf{b}$ is body force per unit mass, such as gravitational acceleration. Decomposing the stress into pressure and deviatoric parts gives

$$\boldsymbol{\sigma} = -p\mathbf{I} + \boldsymbol{\tau}. \tag{7}$$

For a Newtonian fluid,

$$\boldsymbol{\tau} = \mu \left(\nabla \mathbf{v} + (\nabla \mathbf{v})^T \right) + \lambda (\nabla \cdot \mathbf{v}) \mathbf{I}, \tag{8}$$

where μ is dynamic shear viscosity and λ is the second viscosity coefficient. Substitution gives the compressible Navier–Stokes momentum equation for constant viscosities:

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p + \mu \nabla^2 \mathbf{v} + (\mu + \lambda) \nabla (\nabla \cdot \mathbf{v}) + \rho \mathbf{b}. \tag{9}$$

For an incompressible fluid, $\nabla \cdot \mathbf{v} = 0$, so this becomes

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p + \mu \nabla^2 \mathbf{v} + \rho \mathbf{b} + \mathbf{f}_s, \tag{10}$$

where $\mathbf{f}_s$ denotes an optional interfacial force per unit volume, such as surface tension. Angular momentum is conserved when the stress tensor is symmetric, as it is for an ordinary Newtonian fluid without couple stresses.

2.1.4 Pressure Closure

Mass and momentum conservation do not by themselves determine pressure. The closure depends on the SPH formulation:

- **Weakly compressible SPH (WCSPH):** Pressure is computed from an equation of state. A common choice is Tait's equation,

$$p = \frac{\rho_0 c_0^2}{\gamma} \left[\left(\frac{\rho}{\rho_0} \right)^{\gamma} - 1 \right], \tag{11}$$

  where ρ_0 is reference density, c_0 is an artificial numerical speed of sound, and γ controls stiffness. The fluid is allowed a small, controlled density variation.

- **Incompressible or projection SPH (ISPH):** Pressure acts as a Lagrange multiplier enforcing $\nabla \cdot \mathbf{v} = 0$. Taking the divergence of a provisional-velocity correction produces a pressure Poisson equation, schematically

$$\nabla \cdot \left(\frac{1}{\rho} \nabla p \right) = \frac{1}{\Delta t} \nabla \cdot \mathbf{v}^*, \tag{12}$$

  followed by

$$\mathbf{v}^{n+1} = \mathbf{v}^* - \Delta t \frac{\nabla p}{\rho}. \tag{13}$$

2.1.5 Conservation of Energy

An isothermal simulation can omit an explicit energy equation. When temperature, compressible thermodynamics, or heat transfer matters, one common internal-energy form is

$$\rho \frac{De}{Dt} = -p\nabla \cdot \mathbf{v} + \boldsymbol{\tau} : \nabla \mathbf{v} - \nabla \cdot \mathbf{q} + \rho r, \tag{14}$$

where e is specific internal energy, $\mathbf{q}$ is heat flux, and r is volumetric heating per unit mass. Fourier's law closes conductive heat flux:

$$\mathbf{q} = -k\nabla T, \tag{15}$$

where k is thermal conductivity. A caloric equation of state, such as $e = e(\rho, T)$, is then also required.

2.1.6 Initial, Boundary, and Interface Conditions

The differential equations are not a complete simulation problem until they are supplied with:

- initial particle positions and initial fields $\rho(\mathbf{x}, 0)$, $\mathbf{v}(\mathbf{x}, 0)$, and, when needed, $T(\mathbf{x}, 0)$;
- wall conditions, normally no penetration and either no slip, prescribed slip, or a specified wall velocity;
- inlet and outlet conditions, such as prescribed velocity, mass flux, or pressure;
- free-surface conditions, usually ambient normal stress and zero tangential stress when surrounding-gas dynamics are neglected; and
- interface jump conditions for multiphase flow, including surface tension, contact angle, and discontinuous material properties.

2.1.7 Fields and Differential Operators

In these equations:

1. **Functions ($f(\mathbf{x})$, $g(\mathbf{x})$):** These represent continuous fluid properties across space and time, such as the velocity field $\mathbf{v}$, density field ρ, or pressure field p.
2. **Partial differential operators (∇, $\nabla\cdot$, ∇^2):** These describe how fluid properties change across space.

- **Gradient (∇p):** Measures spatial pressure differences, which generate the internal force that pushes fluid from high pressure to low pressure.
- **Divergence ($\nabla \cdot \mathbf{v}$):** Measures the rate at which fluid expands or compresses.
- **Laplacian ($\nabla^2 \mathbf{v}$):** Measures spatial variations in velocity to compute internal viscous (friction) forces between fluid layers.

3 Main Adjustable Parameters in an SPH Simulation

SPH parameters fall into two fundamentally different groups. **Physical parameters** define the fluid and should be determined from the modeled material and nondimensional regime. **Numerical parameters** control how that model is discretized and solved; they should be selected through convergence, stability, and sensitivity studies rather than tuned to produce a preferred visual result.

3.1 Physical and Model Parameters

- **Reference density ρ_0:** Sets the fluid's nominal density and, together with particle volume, determines particle mass. Water near room temperature is commonly modeled with $\rho_0 \approx 1000\ \mathrm{kg\,m^{-3}}$.
- **Viscosity μ or $\nu = \mu/\rho$:** Controls physical momentum diffusion. Its importance relative to inertia is measured by the Reynolds number

$$\mathrm{Re} = \frac{UL}{\nu}. \tag{16}$$

- **Body acceleration b:** Usually gravity $\mathbf{g}$, but it may also include rotating-frame or prescribed accelerations. Gravity–inertia balance is characterized by the Froude number $\mathrm{Fr} = U/\sqrt{gL}$.
- **Surface tension σ:** Controls capillary forces at a free surface or material interface. Its importance can be described by the Weber number $\mathrm{We} = \rho U^2 L/\sigma$.
- **Thermal and multiphase properties:** Thermal conductivity k, heat capacity, phase densities and viscosities, contact angle, and equations of state are adjustable only when the modeled physics requires them.

Here U and L are characteristic velocity and length scales. Matching the relevant nondimensional numbers is often more informative than considering each dimensional parameter independently.

3.2 Resolution, Particles, and Kernel

- **Initial particle spacing Δx:** The primary spatial resolution control. In d dimensions, a uniform equal-mass setup commonly uses

$$m \approx \rho_0 (\Delta x)^d. \tag{17}$$

  Halving Δx increases particle count by roughly 2^d, before the additional cost of smaller time steps is considered.

- **Smoothing length h:** Sets the kernel scale and is usually related to particle spacing by

$$h = \eta \Delta x, \tag{18}$$

  where η is typically of order one but depends on the kernel convention and dimension. Increasing h gives more neighbors and smoother, more robust estimates, but blurs small features and increases cost. Too small an h produces noisy, inconsistent estimates and possible particle disorder.

- **Kernel function and support radius:** Cubic spline, Wendland, and other kernels differ in smoothness, stability, pairing resistance, and computational cost. A kernel may have support h, $2h$, or another multiple of h, so values of h are meaningful only with the stated convention.

- **Target neighbor count:** Closely coupled to $h/\Delta x$ and the kernel. Too few neighbors increases sampling error; too many increases smoothing and work. Variable-smoothing-length methods adjust h_i to keep mass or neighbor count within the support approximately constant.

3.3 Compressibility and Pressure Parameters

- **Numerical speed of sound c_0 in WCSPH:** Determines equation-of-state stiffness. A common design criterion is a numerical Mach number

$$\mathrm{Ma} = \frac{U_{\max}}{c_0} \lesssim 0.1, \tag{19}$$

  which keeps density variation near one percent in many flows. A larger c_0 reduces density error but forces a smaller acoustic time step.

- **Equation-of-state exponent γ and background pressure:** Control the pressure–density response. For water-like WCSPH, $\gamma = 7$ is common, but it is a modeling choice rather than a universal constant. Background pressure can reduce tensile instability but may increase boundary errors or stiffness.

- **Pressure-solver tolerance and iteration limit:** In ISPH, PCISPH, DFSPH, and related methods, tighter density or divergence tolerances improve incompressibility at greater cost. The iteration cap must be high enough that ordinary steps actually reach the tolerance; otherwise it silently becomes the effective accuracy control.

3.4 Viscosity, Stabilization, and Particle Regularization

- **Physical-viscosity discretization:** The chosen Laplacian or pairwise formulation must reproduce the intended μ or ν and behave correctly near walls.

- **Artificial-viscosity coefficients α and β:** Add dissipation to capture shocks and suppress interparticle velocity noise. They are important in compressible SPH but can overdamp low-speed free-surface flow if set unnecessarily high.

- **Density diffusion, such as δ-SPH:** A diffusion coefficient damps high-frequency density and pressure oscillations. Too much diffusion smears real density structure and interfaces.

- **XSPH velocity smoothing, shifting, or transport velocity:** Regularizes the particle distribution. Its strength trades improved particle order and stability against added diffusion or unintended transport, so conservation implications must be checked.

3.5 Time Integration and Stability Controls

- **Time step Δt:** Must resolve acoustic propagation, acceleration, viscosity, and sometimes surface tension. A typical explicit choice is

$$\Delta t = C_{\mathrm{CFL}} \min\left(\frac{h}{c_0 + v_{\mathrm{sig}}}, \sqrt{\frac{h}{a_{\max}}}, \frac{h^2}{\nu} \right), \tag{20}$$

  with an additional capillary restriction when surface tension is strong. Here v_{sig} is a characteristic relative or signal velocity and $a_{\max}$ is maximum particle acceleration.

- **CFL or safety coefficient C_{CFL}:** Scales the stability limits. A smaller value is safer and more accurate but increases the number of steps.

- **Integrator:** Explicit Euler, leapfrog, Verlet, predictor– corrector, and Runge–Kutta methods have different accuracy, storage, and stability properties. Solver tolerances and substepping are additional controls for implicit or iterative schemes.

3.6 Boundary and Interaction Parameters

- **Boundary representation and spacing:** Ghost, dummy, dynamic, or semi-analytical boundary methods require enough boundary layers and spacing compatible with the fluid particles and kernel support.

- **Wall velocity and slip model:** Determine whether the physical condition is no slip, free slip, or partial slip. Penalty-based boundaries may also expose stiffness and damping coefficients.

- **Contact and wetting controls:** Contact angle, adhesion, and fluid–solid coupling coefficients matter in capillary and multiphysics problems and should correspond to measurable physics where possible.

3.7 Practical Parameter-Selection Order

A defensible setup usually follows this order:

1. Set the physical scale, material properties, forcing, and boundary conditions; identify the relevant Reynolds, Froude, Weber, and Mach numbers.

2. Choose Δx from the smallest feature that must be resolved, then choose the kernel and a compatible $h/\Delta x$ and neighbor count.

3. Select the pressure treatment. For WCSPH, choose c_0 from the allowable density variation; for an incompressible solver, choose explicit density or divergence tolerances.

4. Add only the stabilization needed for the observed numerical mode, and separate artificial dissipation from the target physical viscosity.

5. Compute Δt from all applicable stability restrictions rather than selecting it independently.

6. Repeat the simulation at finer Δx, smaller Δt, and tighter solver tolerances. Quantities of interest should approach a stable value; otherwise the result is parameter-dependent rather than demonstrably converged.

4 Discretization and Its Purpose

4.1 What Is Discretization?

Discretization is the mathematical process of converting **continuous** differential equations and infinite spatial fields into a **discrete**, finite form that a digital computer can compute step by step.

In SPH, discretization happens through a two-step pipeline:

1. **Continuous integral approximation:** Continuous functions are smoothed by convolving them with a continuous weighting or smoothing kernel function W.

2. **Particle discretization:** Continuous spatial integrals are replaced by discrete summations over nearby neighboring particles:

4.2 How Kernel Smoothing Produces a Continuous Approximation

4.2.1 Field Quantities

A **field** assigns a physical quantity to every point in space and, usually, time. In general, a field can be written as

$$A = A(\mathbf{x}, t), \tag{21}$$

where $\mathbf{x}$ is a spatial position and t is time. Holding time fixed gives the spatial field $A(\mathbf{x})$. For example, the scalar field $\rho(\mathbf{x}, t)$ assigns one density value to each point, while the vector field $\mathbf{v}(\mathbf{x}, t)$ assigns a velocity vector to each point.

- **Scalar fields** assign one number to each location. Examples include density ρ, pressure p, and temperature T.
- **Vector fields** assign a vector to each location. Examples include velocity $\mathbf{v}$ and force per unit volume $\mathbf{f}$.

In continuum mechanics, these fields are treated as continuous: a value is defined at every point, including the infinitely many points between any two locations. An SPH computer model cannot store infinitely many values. It stores field samples $A_j = A(\mathbf{x}_j)$ only at particle positions $\mathbf{x}_j$ and reconstructs approximate field values between particles.

4.2.2 The Dirac Delta Distribution

The **Dirac delta** $\delta(\mathbf{x} - \mathbf{x}')$ is an idealized point selector. It is called a distribution rather than an ordinary function because it is defined by how it behaves inside an integral. Informally, it is zero everywhere except at $\mathbf{x}' = \mathbf{x}$, is infinitely concentrated at that point, and has unit integral:

$$\delta(\mathbf{x} - \mathbf{x}') = 0 \quad \text{for } \mathbf{x}' \neq \mathbf{x}, \qquad \int_\Omega \delta(\mathbf{x} - \mathbf{x}')\, d\mathbf{x}' = 1, \tag{22}$$

provided that $\mathbf{x}$ lies inside Ω. Its defining **sifting property** is

$$\int_\Omega A(\mathbf{x}')\delta(\mathbf{x} - \mathbf{x}')\, d\mathbf{x}' = A(\mathbf{x}). \tag{23}$$

Here, $\mathbf{x}$ is the location where the field is being evaluated, while $\mathbf{x}'$ is a dummy integration variable that ranges over the domain. The delta eliminates every contribution except the one at $\mathbf{x}' = \mathbf{x}$, so the integral returns the exact field value at $\mathbf{x}$. Thus, the exact value of a field can be written as

$$A(\mathbf{x}) = \int_\Omega A(\mathbf{x}')\delta(\mathbf{x} - \mathbf{x}')\, d\mathbf{x}'. \tag{24}$$

This identity is the starting point for SPH smoothing. SPH replaces the infinitely concentrated Dirac delta with a smooth, finite-width kernel $W(\mathbf{x} - \mathbf{x}', h)$ of finite width. This gives the kernel approximation

$$\langle A(\mathbf{x})\rangle = \int_{\Omega} A(\mathbf{x}')W(\mathbf{x}-\mathbf{x}',h)\,d\mathbf{x}'. \tag{25}$$

This operation is called a **convolution**. To estimate the field at $\mathbf{x}$, it forms a weighted average of values in a neighborhood around $\mathbf{x}$. Nearby points receive greater weight, while points outside the kernel's support receive no weight. A useful smoothing kernel normally has the following properties:

- **Normalization:**
$$\int_{\mathbb{R}^d} W(\mathbf{r},h)\,d\mathbf{r} = 1. \tag{26}$$
This ensures that smoothing reproduces a constant field exactly.

- **Dirac-delta limit:**
$$\lim_{h\to 0} W(\mathbf{r},h) = \delta(\mathbf{r}). \tag{27}$$
More precisely, this limit holds in the distributional sense: $\int A(\mathbf{x}')W(\mathbf{x}-\mathbf{x}',h)\,d\mathbf{x}' \to A(\mathbf{x})$ for a sufficiently regular field A. Thus, the approximation converges toward the original field as the smoothing region shrinks.

- **Positivity:**
$$W(\mathbf{r},h) \geq 0. \tag{28}$$
Nonnegative weights make the approximation a local weighted average and prevent the kernel itself from introducing new extrema through negative contributions.

- **Symmetry:**
$$W(\mathbf{r},h) = W(-\mathbf{r},h). \tag{29}$$
A radial kernel depends only on $\|\mathbf{r}\|$, so particles at equal distances receive equal weights regardless of direction. This also supports pairwise, antisymmetric force calculations in SPH.

- **Compact support:** There is a finite constant κ such that
$$W(\mathbf{r},h) = 0 \quad \text{when} \quad \|\mathbf{r}\| \geq \kappa h. \tag{30}$$
Only nearby particles contribute, making neighbor summations practical.

The **smoothing length** h controls the width of the neighborhood. A larger h includes more neighbors and produces a smoother but less localized estimate. A smaller h preserves finer detail but may produce a noisy estimate if too few particles contribute.

4.2.3 The Cubic Spline Kernel

A common SPH choice is the compactly supported cubic spline, also called the cubic B-spline or M_4 kernel. Let
$$q = \frac{\|\mathbf{r}\|}{h}. \tag{31}$$
In d spatial dimensions, the convention used here is

$$W(\mathbf{r},h)=\sigma_d\begin{cases}6(q^3-q^2)+1, & 0\le q\le \dfrac{1}{2},\\ 2(1-q)^3, & \dfrac{1}{2}<q\le 1,\\ 0, & \text{otherwise,}\end{cases} \tag{32}$$

where the dimension-dependent normalization constant is

$$\sigma_1=\frac{4}{3h},\qquad \sigma_2=\frac{40}{7\pi h^2},\qquad \sigma_3=\frac{8}{\pi h^3}. \tag{33}$$

Some references instead define the same cubic spline with support radius $2h$. The two formulas differ only in the convention used for the smoothing length; the present form has support radius h and matches the slide notation.

This kernel fulfills the desired conditions:

- **Normalization:** For a radial kernel, changing variables from r to $q=r/h$ gives, with $f(q)$ denoting the piecewise polynomial above,

$$\int_{\mathbb{R}^d} W\,d\mathbf{r}=S_d\sigma_d h^d\int_0^1 f(q)q^{d-1}\,dq, \tag{34}$$

  where $S_1=2$, $S_2=2\pi$, and $S_3=4\pi$ are the surface measures of the unit spheres. The needed integrals are

$$\int_0^1 f(q)\,dq=\frac{3}{8},\qquad \int_0^1 qf(q)\,dq=\frac{7}{80},\qquad \int_0^1 q^2f(q)\,dq=\frac{1}{32}. \tag{35}$$

  Substitution of σ_1, σ_2, or σ_3 therefore makes the integral equal to one in the corresponding dimension.

- **Dirac-delta limit:** Each σ_d contains the required factor h^{-d}, preserving unit integral while the support radius h approaches zero. Consequently, all of the kernel's unit mass becomes concentrated at $\mathbf{r}=\mathbf{0}$, $W(\mathbf{r},h)\to\delta(\mathbf{r})$ as $h\to 0$, so convolution with W approaches exact evaluation.

- **Positivity:** Both polynomial pieces of $f(q)$ are nonnegative on their stated intervals, and $\sigma_d>0$. Hence $W(\mathbf{r},h)\ge 0$.

- **Symmetry:** The kernel depends on $\mathbf{r}$ only through $q=\|\mathbf{r}\|/h$. Since $\|-\mathbf{r}\|=\|\mathbf{r}\|$, it follows that $W(-\mathbf{r},h)=W(\mathbf{r},h)$.

- **Compact support:** The kernel is zero outside $q\le 1$ and also reaches zero at $q=1$. Therefore $W(\mathbf{r},h)=0$ whenever $\|\mathbf{r}\|\ge h$, so $\kappa=1$.

To evaluate the integral using particles, SPH assigns particle j the volume element

$$\Delta V_j\approx\frac{m_j}{\rho_j}, \tag{36}$$

where m_j and ρ_j are its mass and density. Replacing the integral by a sum of these particle volumes gives

$$\langle A(\mathbf{x}_i)\rangle \approx \sum_j A_j \frac{m_j}{\rho_j} W(\mathbf{x}_i - \mathbf{x}_j, h) \tag{37}$$

Each term is particle j's field value multiplied by its represented volume and its kernel weight relative to particle i. Although the field data is stored only at discrete particles, W is a continuous function of position. The same sum can therefore be evaluated at any location $\mathbf{x}$, producing a continuous interpolated field from discrete samples.

Similarly, spatial derivatives on fluid particles shift directly onto gradients of the kernel function W.

4.3 Worked PDE Example: The Continuity Equation

Consider conservation of mass in Eulerian form:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \tag{38}$$

Using the material derivative

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla, \tag{39}$$

the same PDE can be written in Lagrangian form as

$$\frac{D\rho}{Dt} = -\rho \nabla \cdot \mathbf{v}. \tag{40}$$

This form is natural for SPH because each particle moves with the fluid.

4.3.1 How SPH Processes the Field Quantities

The continuous fields are replaced by values carried by particles:

- The density field $\rho(\mathbf{x}, t)$ becomes one scalar sample $\rho_i(t)$ on particle i.
- The velocity field $\mathbf{v}(\mathbf{x}, t)$ becomes one vector sample $\mathbf{v}_i(t)$ on particle i.
- The particle position $\mathbf{x}_i(t)$ specifies where those field samples are currently located.
- The mass m_i is usually fixed and represents the amount of fluid carried by the particle. The ratio m_i/ρ_i approximates its volume.

When a field value is required between particles, SPH reconstructs it using the kernel sum. For example,

$$\langle \rho(\mathbf{x})\rangle \approx \sum_j m_j W(\mathbf{x} - \mathbf{x}_j, h). \tag{41}$$

At particle i, this gives the commonly used density summation

$$\rho_i \approx \sum_j m_j W_{ij}, \qquad W_{ij} = W(\mathbf{x}_i - \mathbf{x}_j, h). \tag{42}$$

Thus, field processing means storing samples on particles and using weighted neighbor sums whenever a continuous value must be reconstructed.

4.3.2 How SPH Processes the Differential Operators

SPH replaces each differential operator with an operation on moving particles and kernel derivatives:

- The material derivative D/Dt becomes an ordinary time derivative following particle i. Therefore $D\rho/Dt$ becomes $d\rho_i/dt$.
- The divergence $\nabla \cdot \mathbf{v}$ becomes a weighted sum of velocity differences dotted with kernel gradients:

$$(\nabla \cdot \mathbf{v})_i \approx \frac{1}{\rho_i} \sum_j m_j (\mathbf{v}_j - \mathbf{v}_i) \cdot \nabla_i W_{ij}. \tag{43}$$

The difference $\mathbf{v}_j - \mathbf{v}_i$ is important: if all neighboring particles have the same velocity, every difference is zero and SPH correctly estimates zero divergence.

For the cubic spline used above, let $\mathbf{r}_{ij} = \mathbf{x}_i - \mathbf{x}_j$, $r_{ij} = \|\mathbf{r}_{ij}\|$, and $q = r_{ij}/h$. For $r_{ij} > 0$, its gradient is

$$\nabla_i W_{ij} = \frac{\sigma_d}{h} f'(q) \frac{\mathbf{r}_{ij}}{r_{ij}}, \tag{44}$$

where

$$f'(q) = \begin{cases} 18q^2 - 12q, & 0 \le q \le \frac{1}{2}, \\ -6(1-q)^2, & \frac{1}{2} < q \le 1, \\ 0, & \text{otherwise.} \end{cases} \tag{45}$$

At $r_{ij} = 0$, symmetry gives $\nabla_i W_{ij} = \mathbf{0}$.

4.3.3 The Resulting Particle Equation

Substituting the SPH divergence approximation into the continuity equation gives

$$\frac{d\rho_i}{dt} = -\rho_i (\nabla \cdot \mathbf{v})_i \tag{46}$$

$$\approx \sum_j m_j (\mathbf{v}_i - \mathbf{v}_j) \cdot \nabla_i W_{ij}. \tag{47}$$

The continuous PDE has now become one ordinary differential equation for each particle. A simulation step processes it as follows:

1. Find all neighbors j satisfying $\|\mathbf{x}_i - \mathbf{x}_j\| < h$.
2. Read the sampled field values ρ_i, $\mathbf{v}_i$, and $\mathbf{v}_j$ from the particles.
3. Evaluate $\nabla_i W_{ij}$ from particle separation using the cubic spline derivative.
4. Sum the neighbor contributions to obtain $d\rho_i/dt$.
5. Advance density in time; for example, an explicit Euler step is

$$\rho_i^{n+1} = \rho_i^n + \Delta t \left(\frac{d\rho_i}{dt} \right)^n. \tag{48}$$

In summary, SPH handles *field quantities* as particle-carried samples and kernel-weighted reconstructions. It handles *differential operators* by differentiating the smooth kernel and summing the resulting neighbor contributions. This converts the original spatial PDE into a system of time-dependent particle equations.

5 Major SPH Papers and Further Reading

The following papers provide a chronological path from the original astrophysical formulation of SPH to modern incompressible-flow and computer graphics methods. DOI links lead to the publisher record; access to the full text may depend on the reader's institution.

1. **L. B. Lucy (1977), "A Numerical Approach to the Testing of the Fission Hypothesis."** One of the two papers that independently introduced SPH, originally for three-dimensional astrophysical problems. Read via DOI.

2. **R. A. Gingold and J. J. Monaghan (1977), "Smoothed Particle Hydrodynamics: Theory and Application to Non-Spherical Stars."** The other foundational SPH paper; it develops the kernel-based particle formulation and applies it to stellar dynamics. Read via DOI.

3. **J. J. Monaghan (1992), "Smoothed Particle Hydrodynamics."** A classic review that systematizes the mathematical formulation, kernels, conservation properties, artificial viscosity, and early astrophysical applications. Read via DOI.

4. **J. P. Morris, P. J. Fox, and Y. Zhu (1997), "Modeling Low Reynolds Number Incompressible Flows Using SPH."** An influential treatment of physical viscosity and wall boundaries for viscous incompressible flows. Read via DOI.

5. **S. J. Cummins and M. Rudman (1999), "An SPH Projection Method."** Introduces a pressure-projection approach that enforces incompressibility through a pressure Poisson equation rather than a stiff equation of state. Read via DOI.

6. **M. Müller, D. Charypar, and M. Gross (2003), "Particle-Based Fluid Simulation for Interactive Applications."** A landmark computer-graphics paper that presents an efficient SPH formulation for interactive liquid simulation, including pressure, viscosity, and surface tension. Read via DOI.

7. **J. J. Monaghan (2005), "Smoothed Particle Hydrodynamics."** A broad and more modern review of SPH theory, numerical behavior, and applications across astrophysics and continuum mechanics. Read via DOI.

8. **M. Becker and M. Teschner (2007), "Weakly Compressible SPH for Free Surface Flows."** A widely used graphics formulation that controls density variation with a stiff equation of state and is designed for efficient free-surface animation. Read via DOI.

9. **B. Solenthaler and R. Pajarola (2009), "Predictive-Corrective Incompressible SPH."** Introduces PCISPH, which iteratively predicts and corrects pressure to reduce density error while allowing larger time steps than many weakly compressible formulations. Read via DOI.

10. **D. J. Price (2012), "Smoothed Particle Hydrodynamics and Magnetohydrodynamics."** A detailed derivation-focused review covering consistency, conservation, shocks, dissipa-

tion, and magnetohydrodynamics; it is useful even when the main interest is ordinary fluid SPH. Read via DOI.

11. **A. Adami, X. Y. Hu, and N. A. Adams (2012), "A Generalized Wall Boundary Condition for Smoothed Particle Hydrodynamics."** Develops a robust boundary-particle treatment for walls under complex and time-dependent flows. Read via DOI.

12. **M. Ihmsen, J. Orthmann, B. Solenthaler, A. Kolb, and M. Teschner (2014), "SPH Fluids in Computer Graphics."** A useful survey of the complete SPH simulation pipeline, including neighborhood search, incompressibility solvers, boundary handling, surface reconstruction, and implementation considerations. Read via DOI.